

10th lesson work – Thermal Physics

15/15

Please see below the assigned work for the 10th lesson. This should take you approximately 1 hour and is designed to be completed in your physics study period – the allocated hour in your timetable with no teacher. You may choose to do this work digitally or on paper; in school or remotely.

The answers are included, for you to mark your own work. You are expected to show evidence of completion eg workings, as well as submitting your final score out of **15**. Please attempt all questions before going ahead to the answers (you will get more out of it if you do!).

1. The freezing point of ethanol is 159 K.
What is 159 K in °C?

- A – 432 °C
- B – 114 °C
- C 114 °C
- D 432 °C

$$\begin{aligned} -273^{\circ}\text{C} &= 0\text{K} \\ \Rightarrow 159\text{K} &= -273 + 159 \\ &= -114^{\circ}\text{C} \end{aligned}$$

Your answer

B

[1]

2.

Define the *internal energy* of a substance.

The sum of all the thermal energy
(kinetic energy of the atoms) and the sum
of the chemical potential energies.

More simply Sum KE + Sum PE [1]

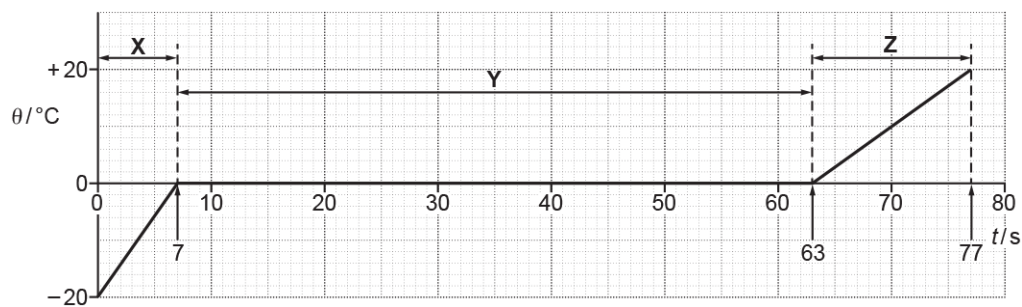
3. Brownian motion provides evidence for a kinetic model of gases. State **three** key assumptions made in the kinetic theory of gases.

- Net Motion of particles are random (no preferred direction)
- Time of collision is negligible compared to time between collisions.
- Volume of particle negligible compared to volume of container

seems correct but not on MS?

[3]

4. A 150 W heater is used to heat 25 g of ice in a sealed and well-insulated container. The initial temperature of the ice is -20 °C. The graph shows the variation of temperature θ with time t as the ice is heated.



There are three distinct regions of the graph, **X**, **Y** and **Z**.

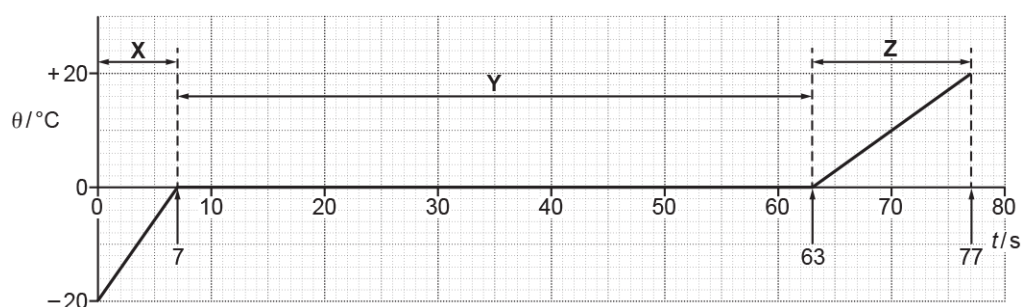
- i. Describe the motions of the molecules in region **X** and in region **Z**.

In Region X the molecules are fixed in a lattice (it's a solid).
In region Z the molecules are free to flow (it's a liquid)

[2]

- ii. The internal energy of the ice increases from $t = 0$ to $t = 77$ s.
Complete the table below using the following key for the physical quantities:

- K = kinetic energy of molecules
- P = potential energy of molecules.



Region	Physical quantity, or quantities, that increases as time increases	Physical quantity, or quantities, that remain constant as time increases
X	kinetic energy	Potential energies
Y	Potential energies	kinetic energy
Z	kinetic energy	Potential energies

[3]

- iii. State the temperature of the ice at which its molecules have zero kinetic energy.

-273°C

[1]

can also just say 0K

5. A substance can exist as a crystalline solid, a liquid or a gas. A solid sample of the substance is placed in a sealed container and heated at a constant rate until it changes into a gas.

Fig. 21 shows the variation with time t of the temperature θ for the substance.

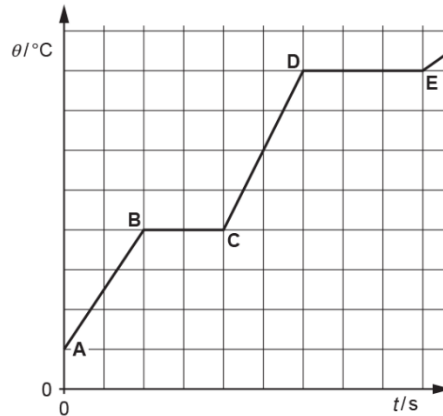


Fig. 21

Use the kinetic theory of matter to describe the solid phase (section **AB**) and the liquid phase (section **CD**) in terms of the motion and arrangement of the molecules of the substance.

Section

AB:

The molecules are vibrating around a fixed point. The molecules are arranged in a regular lattice.

Section

CD:

The molecules are free to flow and are arranged in a non-regular pattern.

Mark scheme

Question			Answer/Indicative content	Marks	Guidance
1			B	1	
			Total	1	
2			The sum of (the random distribution of) the KE and PE of (its) molecules	B1	Not if no clear indication of particulate nature, i.e. allow particles or atoms for molecules Examiner's Comments The correct answer for this item was a direct reference to specification point 5.1.2 (d) and required the association with the particles of a system. Many more than half of the candidates would have scored this mark had they included this association.
			Total	1	
3			Any three from: - Forces between particles are negligible except during collisions - Collisions are perfectly elastic - Time of a collision is negligible compared to time between collision - Particles / atoms / molecules occupy negligible volume compared to volume of gas - Large number of molecules in random motion.	B1 × 3	
			Total	3	
4		i	Molecules in X vibrate about fixed positions /AW Molecules in Z are free to move/random/AW	B1 B1	Allow references to ice for X and water / liquid for Z Allow one correct for B1 from: Molecules in X have lower <u>KE</u>/speed/velocity Speed/velocity of molecules increases with temp/time

					Amplitude or frequency increases with temp/time in X
		ii	Region Physical quantity, or quantities, that increase as time increases Physical quantity, or quantities, that remain constant as time increases X K P Y P K Z K P	B1×3	Note that each B1 mark is for a correct row Allow KP/- for both X and Z
		iii	Absolute zero / 0 <u>K</u> / - 273 <u>°C</u>	B1	
			Total	6	
5			Section AB Any <u>two</u> from • Particles close together • Particle spacing increase with increasing time or increasing temperature • Particles in a fixed structure/(regular) lattice • Particles vibrate/perform SHM • Particles vibrate with increasing amplitude (from A to B) Section CD Any <u>two</u> from • Particles close together //(slightly) further apart (than in AB) • No regular structure /AW • Particles (are free to) move around / move past each other / flow • Particles move with increasing speed from C to D / greater KE	B1 x 2 B1 x 2	Not: 'vibrates more'
			Total	4	